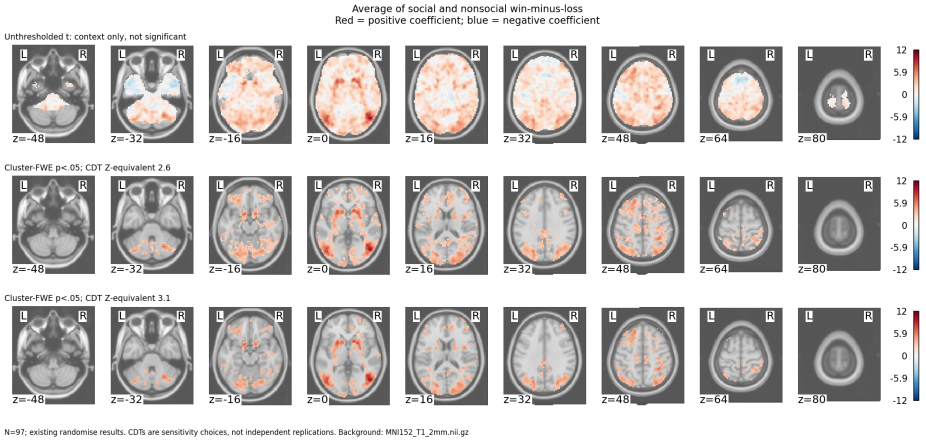
Win > Loss: Expected Striatal Activation Is Present

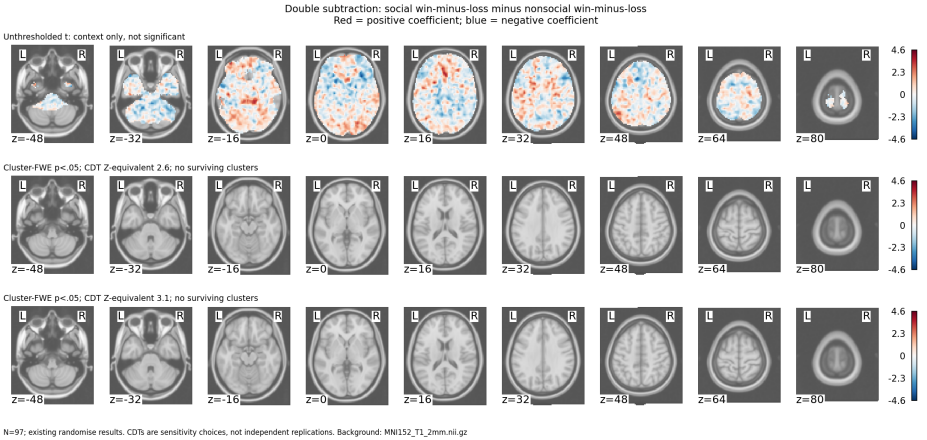
PRELIMINARY | Frozen N=97 | Exploratory, not preregistered

Context

Yes: both Social Doors and nonsocial Doors show bilateral ventral-striatal-region win > loss responses. The across-task average confirms the expected reward response; social specificity is a separate question.

Content





Outcome	Existing result	Interpretation
Average win > loss across domains	Minimum cluster-FWE p=.0001 at both CDTs; 11,811 corrected voxels at CDT 3.1	Bilateral ventral-striatal-region peaks: t=11.79 left, 11.70 right (left figure).
Double subtraction: (social win-loss) - (nonsocial win-loss)	No corrected cluster in either direction at either CDT	No detected group-average domain-by-valence effect (right figure).
Social win - nonsocial win	Corrected positive and negative clusters at both CDTs; see next page	A domain difference in winning responses, not a win-loss interaction.

Conclusion

The expected striatal reward response is present, not missing: averaged win > loss peaks near MNI [-12.7, 6.3, -9.6] and [14.3, 9.0, -9.6] lie in corrected clusters with FWE p=.0001. The null double subtraction means we do not detect a group-average social advantage in win-loss differentiation.

Figure rows: unthresholded context, corrected CDT 2.6, corrected CDT 3.1. Red/blue show positive/negative coefficients. All results use existing no-TEDANA sm-5 FEAT inputs; no preprocessing or exclusions changed.

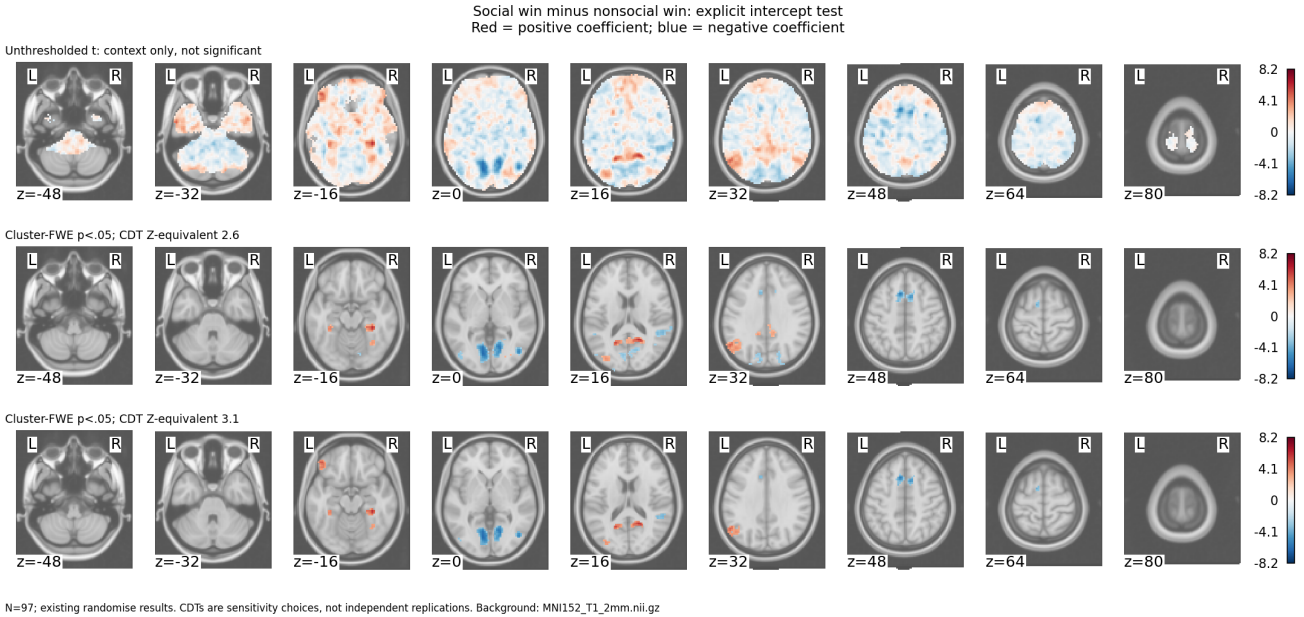
Group Mean: Social Win Minus Nonsocial Win

PRELIMINARY | Frozen N=97 | Exploratory, not preregistered

Context

Does the response to winning differ between social and nonsocial contexts? This is distinct from the double subtraction, which additionally references losses.

Content



N=97; existing randomise results. CDTs are sensitivity choices, not independent replications. Background: MNI152_T1_2mm.nii.gz

CDT (Z-equivalent)	Direction	Minimum cluster-FWE p	Corrected voxels
2.6	positive	0.0175	1196
2.6	negative	0.0019	1991
3.1	positive	0.0021	762
3.1	negative	0.0011	1008

Conclusion

There are regions with social > nonsocial winning responses and regions showing the reverse, at both thresholds. This establishes domain differentiation for winning, not a universal social-reward increase or a domain-by-valence interaction. It should not be presented as proof of additional striatal activation for social wins.

The displayed domain effects include temporal and posterior cortical areas, with both positive and negative regions. These are coarse visual descriptions, not new atlas assignments; exact striatal overlap for this contrast has not been quantified.

Both new intercept jobs and four directional maps passed validation, including the native sign-flip gate; 10,000 permutations, residual df=90. The pre-existing win-loss and double-subtraction tests were reused, not rerun.

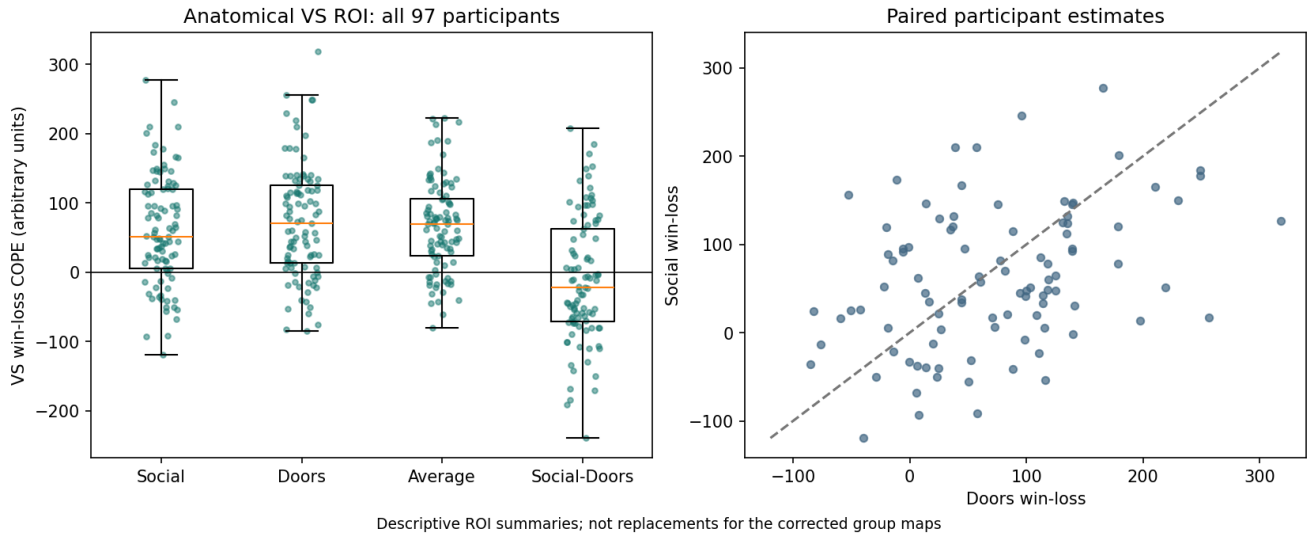
Striatal Sanity Check: Both Tasks Work

PRELIMINARY | Frozen N=97 | Exploratory, not preregistered

Context

Is the win > loss signal actually present in striatum, rather than merely somewhere in a large whole-brain map? Existing bilateral cluster peaks and the pre-existing anatomical VS ROI both support it.

Content



Descriptive ROI summaries; not replacements for the corrected group maps

Win > loss outcome	Left / right regional peak t	Left / right cluster-FWE p	Anatomical VS mean [descriptive 95% CI]
Social Doors	7.56 / 8.96	0.0007 / 0.0001	60.98 [45.02, 76.94]
Nonsocial Doors	9.85 / 10.78	0.0003 / 0.0003	72.28 [55.77, 88.78]
Across-task average	11.79 / 11.70	0.0001 / 0.0001	66.63 [53.24, 80.02]

ROI values are arbitrary COPE units, not percent signal change. The anatomical seed-VS mask was intersected with the common analysis mask, and all 97 participants remain. These are descriptive ROI summaries, not new small-volume-corrected tests. Peak p-values are for the containing cluster at CDT 3.1, not individual-voxel FWE claims or formal anatomical-nucleus labels.

Conclusion

Yes, we recover the expected bilateral striatal-region win > loss response in both tasks. The anatomical VS social-minus-nonsocial win-loss mean is -11.29 [descriptive 95% CI -29.65, 7.06], consistent with the null corrected double subtraction. That quantity is not the social-win-minus-nonsocial-win contrast, because it also subtracts losses.

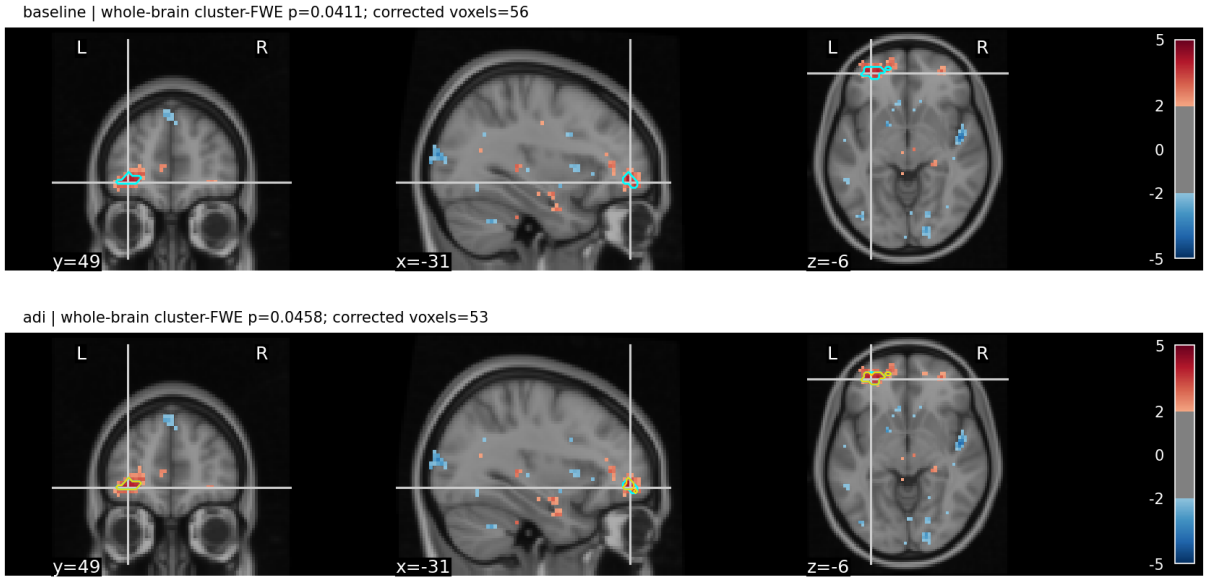
FEVS And Social-Winning Activation: Original / +ADI

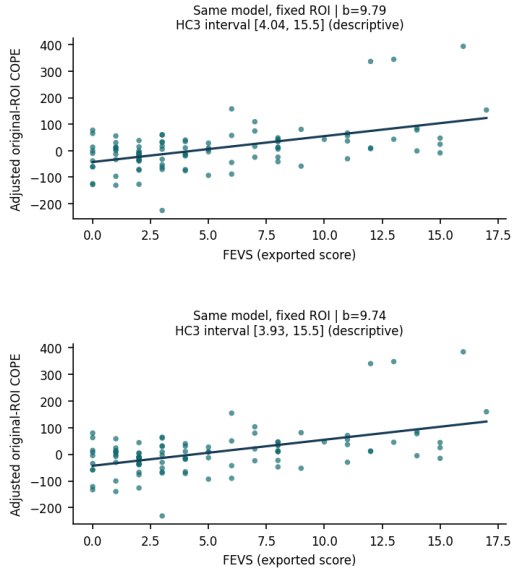
PRELIMINARY | Frozen N=97 | Exploratory, not preregistered

Context

Does financial vulnerability relate to the social win-minus-nonsocial win activation contrast?

Content





Each brain row is beside its exact model-adjusted scatter. Cyan: original corrected cluster; gold: adjusted corrected cluster. The same original ROI is used for every scatter, including whole-brain null fits. Brain shading |t|>=2 is descriptive, not corrected significance.

Model	Cluster-FWE p, CDT 3.1	ROI slope / HC3 interval (descriptive)	Leave-one-out slope range / max Cook's D
baseline	0.0411	9.79 [4.04, 15.5]	[7.86, 10.4] / 0.206
adi	0.0458	9.74 [3.93, 15.5]	[7.85, 10.4] / 0.294

Conclusion

The original association is positive, but adding Ecog removes whole-brain corrected significance. Treat this as qualified evidence, not an adjustment-robust cortical vmPFC/OFC finding.

Anatomy caveat: original clusters are predominantly white-matter-labeled in a standard atlas; the jointly adjusted conditional cluster is approximately 64% white-matter-labeled. The conditional activation centroid shifts only 1.1 mm with joint adjustment. Neither fact validates a clean cortical vmPFC/OFC claim.

All 97 participants retained. Selected-ROI scatter slopes/HC3 intervals are descriptive only, not independent confirmation or a rescue of a null whole-brain result. No cases removed; leave-one-out checks do not exclude collective influence. Findings pass only at CDT 3.1; correction is per map, not across the exploratory model family. None of these focal models contains MSPSS.

FEVS And Social-Winning Activation: +Ecog / +Both

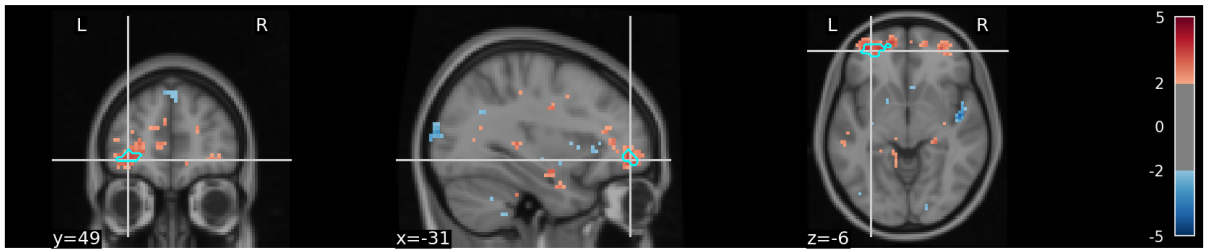
PRELIMINARY | Frozen N=97 | Exploratory, not preregistered

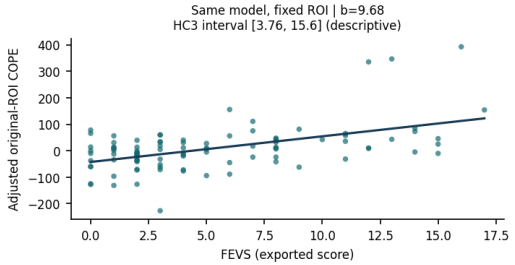
Context

Does financial vulnerability relate to the social win-minus-nonsocial win activation contrast?

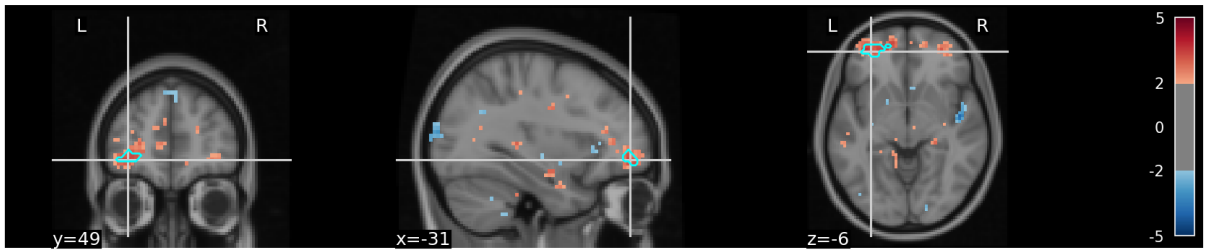
Content

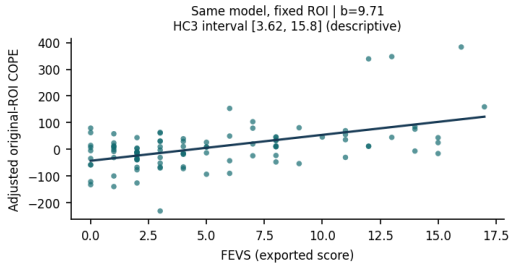
ecog | whole-brain cluster-FWE p=0.0666; corrected voxels=0





adi_ecog | whole-brain cluster-FWE p=0.0709; corrected voxels=0





Each brain row is beside its exact model-adjusted scatter. Cyan: original corrected cluster; gold: adjusted corrected cluster. The same original ROI is used for every scatter, including whole-brain null fits. Brain shading |t|>=2 is descriptive, not corrected significance.

Model	Cluster-FWE p, CDT 3.1	ROI slope / HC3 interval (descriptive)	Leave-one-out slope range / max Cook's D
ecog	0.0666	9.68 [3.76, 15.6]	[7.93, 10.3] / 0.214
adi_ecog	0.0709	9.71 [3.62, 15.8]	[7.92, 10.4] / 0.264

Conclusion

The original association is positive, but adding Ecog removes whole-brain corrected significance. Treat this as qualified evidence, not an adjustment-robust cortical vmPFC/OFC finding.

Anatomy caveat: original clusters are predominantly white-matter-labeled in a standard atlas; the jointly adjusted conditional cluster is approximately 64% white-matter-labeled. The conditional activation centroid shifts only 1.1 mm with joint adjustment. Neither fact validates a clean cortical vmPFC/OFC claim.

All 97 participants retained. Selected-ROI scatter slopes/HC3 intervals are descriptive only, not independent confirmation or a rescue of a null whole-brain result. No cases removed; leave-one-out checks do not exclude collective influence. Findings pass only at CDT 3.1; correction is per map, not across the exploratory model family. None of these focal models contains MSPSS.

Activation With PM2.5 In The Model: Original / +ADI

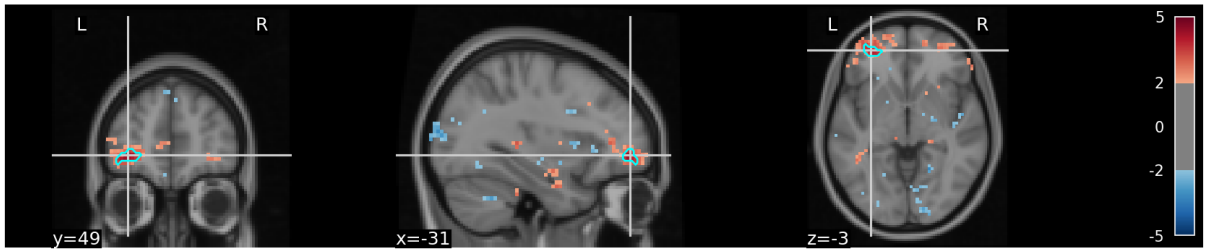
PRELIMINARY | Frozen N=97 | Exploratory, not preregistered

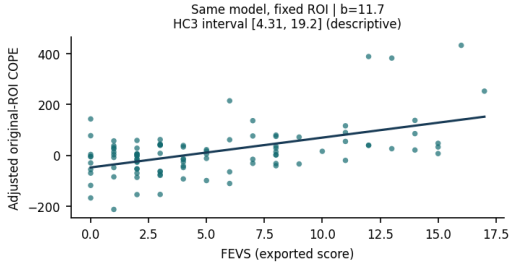
Context

Does the FEVS activation coefficient remain when PM2.5 and FEVS x PM2.5 are included? The coefficient is evaluated at mean PM2.5.

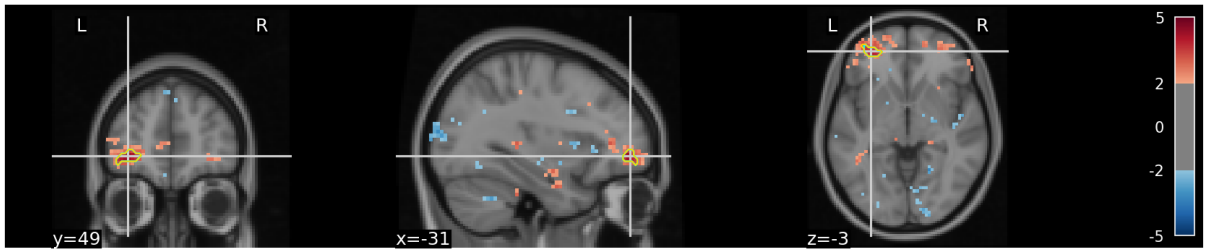
Content

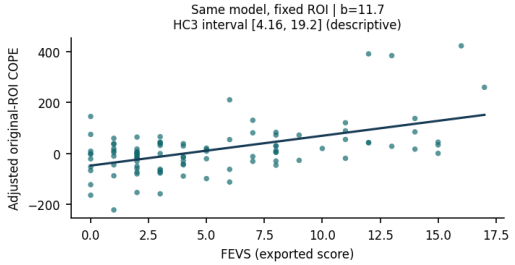
baseline | whole-brain cluster-FWE p=0.0315; corrected voxels=67





adi | whole-brain cluster-FWE p=0.0323; corrected voxels=66





Each brain row is beside its exact model-adjusted scatter. Cyan: original corrected cluster; gold: adjusted corrected cluster. The same original ROI is used for every scatter, including whole-brain null fits. Brain shading |t|>=2 is descriptive, not corrected significance.

Model	Cluster-FWE p, CDT 3.1	ROI slope / HC3 interval (descriptive)	Leave-one-out slope range / max Cook's D
baseline	0.0315	11.7 [4.31, 19.2]	[9.7, 12.7] / 0.255
adi	0.0323	11.7 [4.16, 19.2]	[9.63, 12.6] / 0.322

Conclusion

The conditional FEVS coefficient retains a similar location, but correction is borderline and depends on adjustment. This is a main coefficient, not a significant winning-related FEVS x PM2.5 interaction.

Anatomy caveat: original clusters are predominantly white-matter-labeled in a standard atlas; the jointly adjusted conditional cluster is approximately 64% white-matter-labeled. The conditional activation centroid shifts only 1.1 mm with joint adjustment. Neither fact validates a clean cortical vmPFC/OFC claim.

All 97 participants retained. Selected-ROI scatter slopes/HC3 intervals are descriptive only, not independent confirmation or a rescue of a null whole-brain result. No cases removed; leave-one-out checks do not exclude collective influence. Findings pass only at CDT 3.1; correction is per map, not across the exploratory model family. None of these focal models contains MSPSS.

Activation With PM2.5 In The Model: +Ecog / +Both

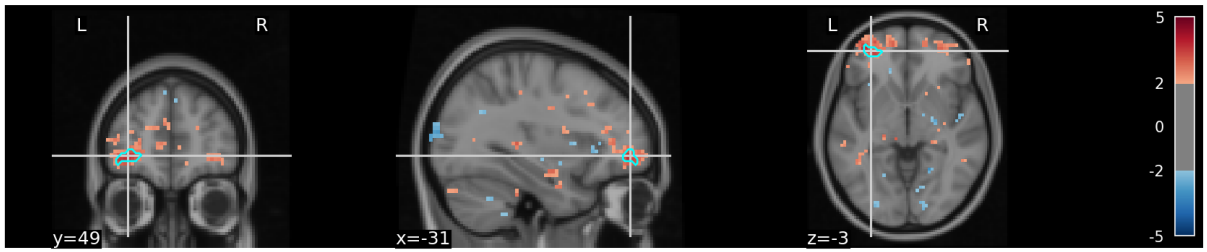
PRELIMINARY | Frozen N=97 | Exploratory, not preregistered

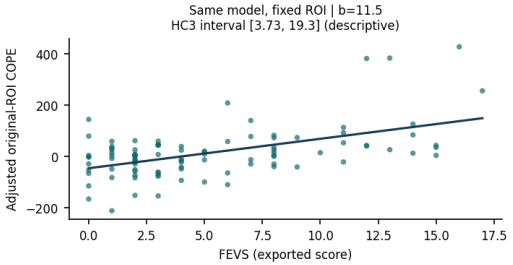
Context

Does the FEVS activation coefficient remain when PM2.5 and FEVS x PM2.5 are included? The coefficient is evaluated at mean PM2.5.

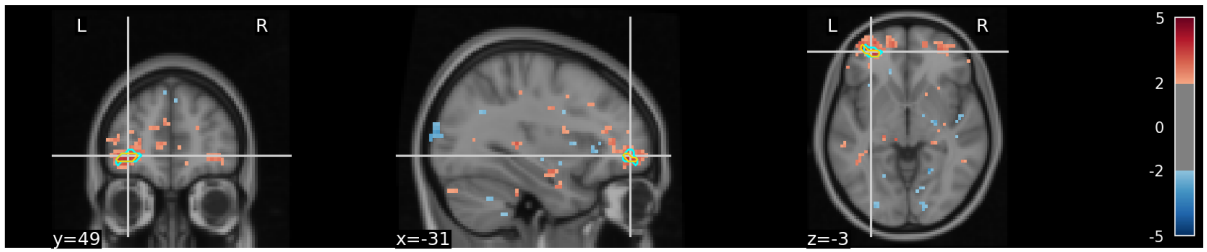
Content

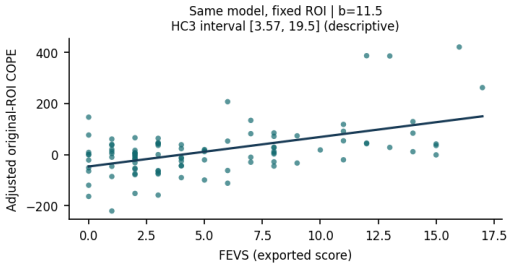
ecog | whole-brain cluster-FWE p=0.0505; corrected voxels=0





adi_ecog | whole-brain cluster-FWE p=0.0477; corrected voxels=53





Each brain row is beside its exact model-adjusted scatter. Cyan: original corrected cluster; gold: adjusted corrected cluster. The same original ROI is used for every scatter, including whole-brain null fits. Brain shading |t|>=2 is descriptive, not corrected significance.

Model	Cluster-FWE p, CDT 3.1	ROI slope / HC3 interval (descriptive)	Leave-one-out slope range / max Cook's D
ecog	0.0505	11.5 [3.73, 19.3]	[9.01, 12.4] / 0.359
adi_ecog	0.0477	11.5 [3.57, 19.5]	[9.03, 12.5] / 0.404

Conclusion

The conditional FEVS coefficient retains a similar location, but correction is borderline and depends on adjustment. This is a main coefficient, not a significant winning-related FEVS x PM2.5 interaction.

Anatomy caveat: original clusters are predominantly white-matter-labeled in a standard atlas; the jointly adjusted conditional cluster is approximately 64% white-matter-labeled. The conditional activation centroid shifts only 1.1 mm with joint adjustment. Neither fact validates a clean cortical vmPFC/OFC claim.

All 97 participants retained. Selected-ROI scatter slopes/HC3 intervals are descriptive only, not independent confirmation or a rescue of a null whole-brain result. No cases removed; leave-one-out checks do not exclude collective influence. Findings pass only at CDT 3.1; correction is per map, not across the exploratory model family. None of these focal models contains MSPSS.

FEVS And Social-By-Valence VS Connectivity: Original / +ADI

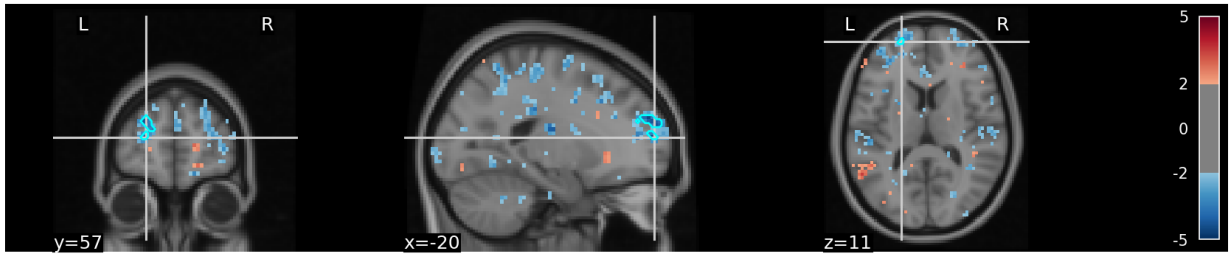
PRELIMINARY | Frozen N=97 | Exploratory, not preregistered

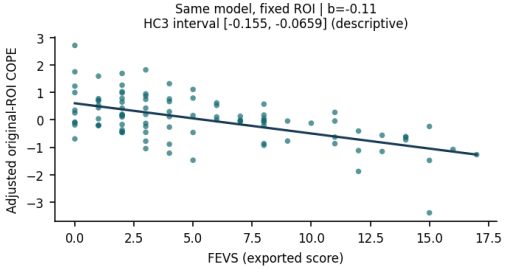
Context

Does financial vulnerability relate to VS coupling for (social win-loss) minus (nonsocial win-loss), at mean PM2.5?

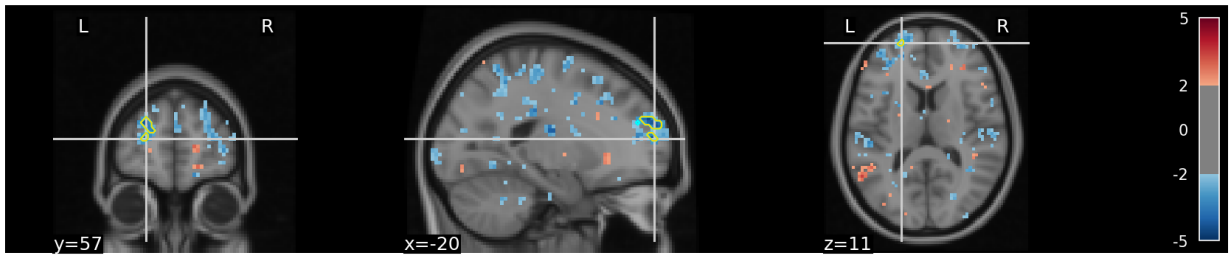
Content

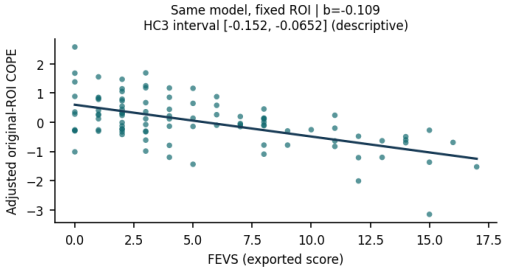
baseline | whole-brain cluster-FWE p=0.0303; corrected voxels=44





adi | whole-brain cluster-FWE p=0.0320; corrected voxels=43





Each brain row is beside its exact model-adjusted scatter. Cyan: original corrected cluster; gold: adjusted corrected cluster. The same original ROI is used for every scatter, including whole-brain null fits. Brain shading |t|>=2 is descriptive, not corrected significance.

Model	Cluster-FWE p, CDT 3.1	ROI slope / HC3 interval (descriptive)	Leave-one-out slope range / max Cook's D
baseline	0.0303	-0.11 [-0.155, -0.0659]	[-0.114, -0.0975] / 0.200
adi	0.0320	-0.109 [-0.152, -0.0652]	[-0.114, -0.0973] / 0.187

Conclusion

Higher FEVS relates to a lower social-by-valence connectivity contrast with a left frontal target. This is more consistent across ADI/Ecog adjustment than activation, but it is not evidence of globally weaker connectivity or a winning-only effect.

The left frontal target is not the VS seed. Joint adjustment retains 42 voxels, 32 overlapping the original 44; centroid shift=1.3 mm. Every adjusted peak remains inside the original cluster.

All 97 participants retained. Selected-ROI scatter slopes/HC3 intervals are descriptive only, not independent confirmation or a rescue of a null whole-brain result. No cases removed; leave-one-out checks do not exclude collective influence. Findings pass only at CDT 3.1; correction is per map, not across the exploratory model family. None of these focal models contains MSPSS.

FEVS And Social-By-Valence VS Connectivity: +Ecog / +Both

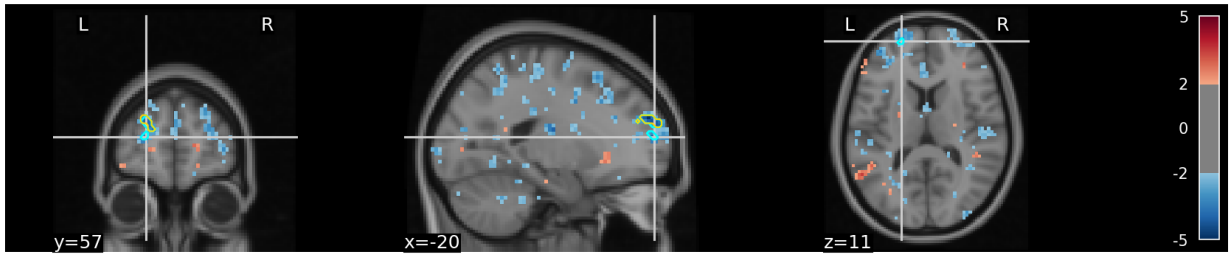
PRELIMINARY | Frozen N=97 | Exploratory, not preregistered

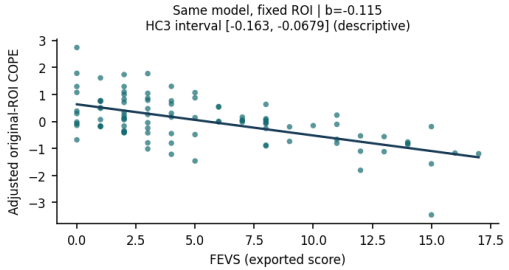
Context

Does financial vulnerability relate to VS coupling for (social win-loss) minus (nonsocial win-loss), at mean PM2.5?

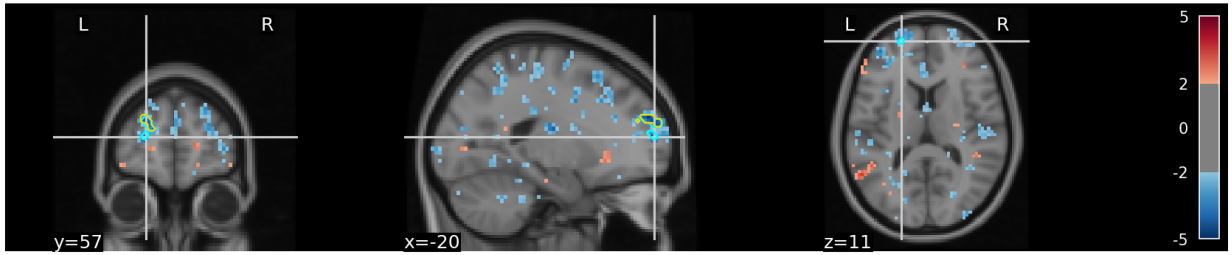
Content

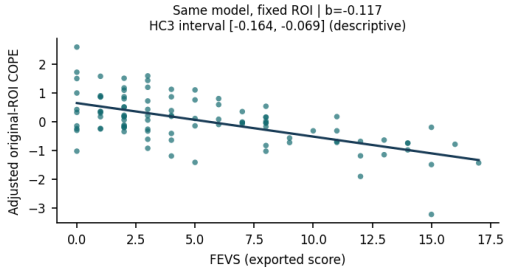
ecog | whole-brain cluster-FWE p=0.0349; corrected voxels=42





adi_ecog | whole-brain cluster-FWE p=0.0353; corrected voxels=42





Each brain row is beside its exact model-adjusted scatter. Cyan: original corrected cluster; gold: adjusted corrected cluster. The same original ROI is used for every scatter, including whole-brain null fits. Brain shading |t|>=2 is descriptive, not corrected significance.

Model	Cluster-FWE p, CDT 3.1	ROI slope / HC3 interval (descriptive)	Leave-one-out slope range / max Cook's D
ecog	0.0349	-0.115 [-0.163, -0.0679]	[-0.122, -0.103] / 0.186
adi_ecog	0.0353	-0.117 [-0.164, -0.069]	[-0.123, -0.105] / 0.173

Conclusion

Higher FEVS relates to a lower social-by-valence connectivity contrast with a left frontal target. This is more consistent across ADI/Ecog adjustment than activation, but it is not evidence of globally weaker connectivity or a winning-only effect.

The left frontal target is not the VS seed. Joint adjustment retains 42 voxels, 32 overlapping the original 44; centroid shift=1.3 mm. Every adjusted peak remains inside the original cluster.

All 97 participants retained. Selected-ROI scatter slopes/HC3 intervals are descriptive only, not independent confirmation or a rescue of a null whole-brain result. No cases removed; leave-one-out checks do not exclude collective influence. Findings pass only at CDT 3.1; correction is per map, not across the exploratory model family. None of these focal models contains MSPSS.

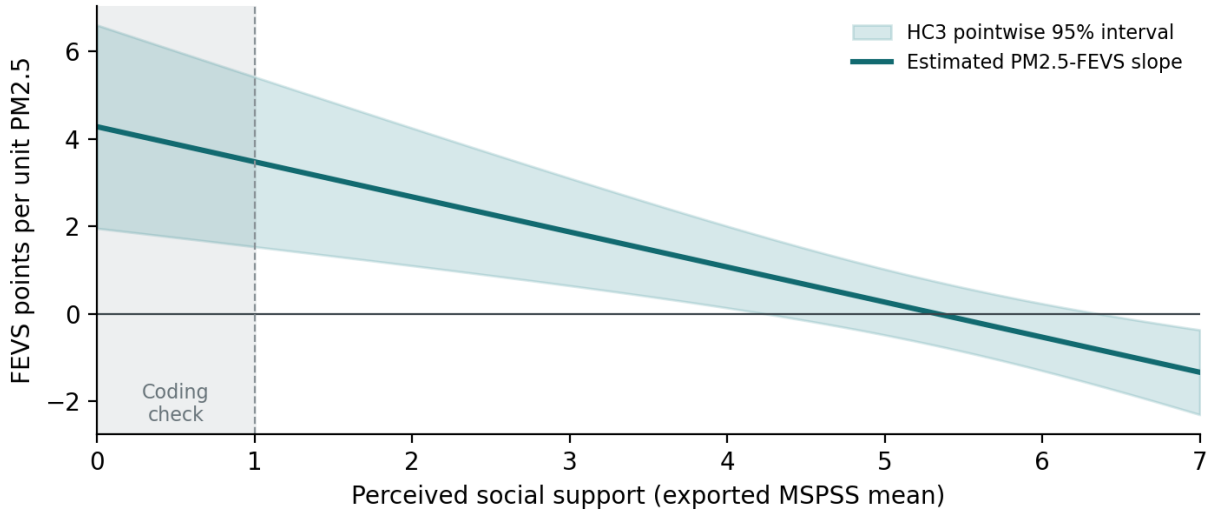
Behavioral Context: Support Result Awaiting Correction

PRELIMINARY | Frozen N=97 | Exploratory, not preregistered

Context

Does perceived support moderate the PM2.5-FEVS association? The previous screener MSPSS coding is now known to be incorrect, so this is a historical result awaiting rerun.

Content



Historical fit	Interaction b	HC3 p (uncorrected sensitivity check)
baseline	-0.802	0.000168
adi	-0.802	0.000224
ecog	-0.714	0.001093
adi+ecog	-0.711	0.001313

The displayed conditional-slope panels use the old scores. Pointwise intervals and any JN regions are exploratory, not simultaneous bands. Original primary behavioral interaction: b=-.802, HC3 CI [-1.208,-.396], p=.000168; Holm p=.000336 across the original two primary tests. Those numbers are not scoring-corrected findings.

Conclusion

Do not finalize the support-buffering story yet. Obtain corrected screener and pre-scan scores/dates, verify scoring, and assess agreement. Preserve N=97; rerun only MSPSS-dependent models and their existing multiplicity summaries after verification.

The focal FEVS neural models shown earlier do not include MSPSS and are unaffected by its correction. No matching corrected neural PM2.5 x support result has been established. Separate behavioral and neural associations do not demonstrate mediation.

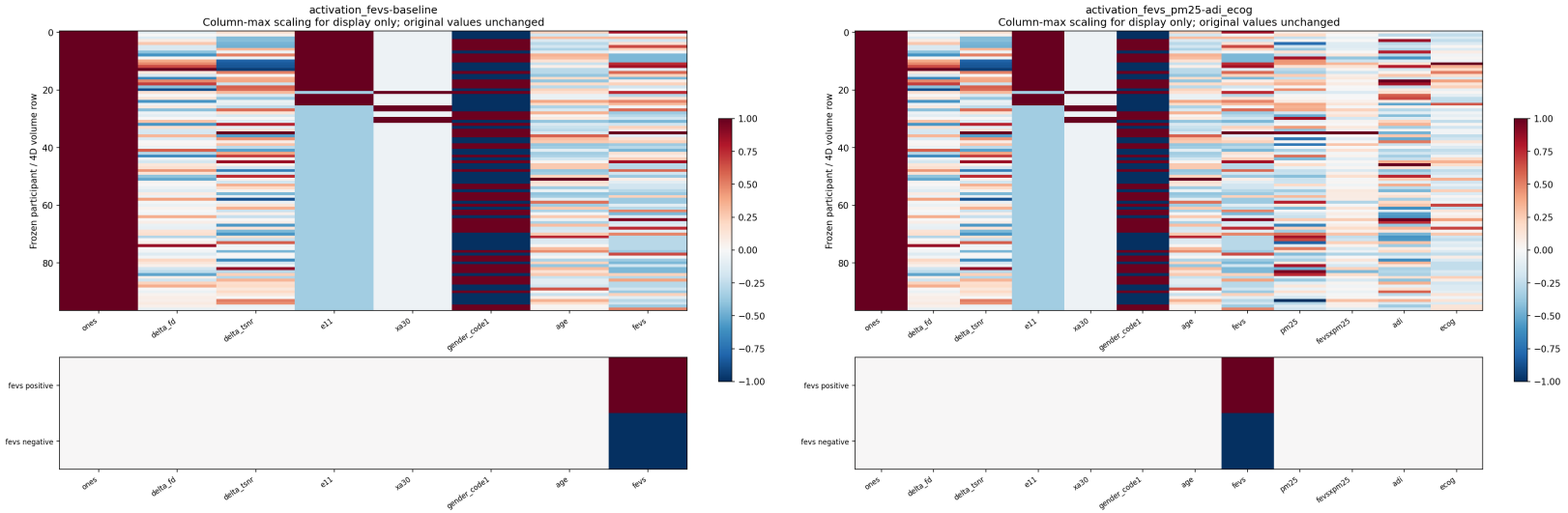
Design Audit And Reporting Decisions

PRELIMINARY | Frozen N=97 | Exploratory, not preregistered

Context

Were the intended coefficients actually contrasted, and does each design retain the same 97 image-volume rows?

Content



Left: standalone FEVS. Right: FEVS/PM2.5 with additive ADI and Ecog. These published heatmaps read actual executed .mat/.con files, not newly chosen designs, and are not native FEAT screenshots. Display-only column scaling; participant identifiers omitted. The separate complete audit verified image/design ordering.

Executed model	Rank / residual df	Condition number (scaled)	Intercept tested?
activation_fevs-baseline	8 / 89	2.14	No
activation_fevs_pm25-baseline	10 / 87	2.23	No
activation_fevs_pm25-adi_ecog	12 / 85	2.39	No
task-mean-intercept	7 / 90	2.14	Yes

Check	Verdict
Column of ones	Included, but NOT tested in the plotted FEVS +/- contrasts. An intercept must have its own nonzero contrast weight.
FEVS coefficient versus interaction	Standalone plot tests FEVS. PM2.5-containing figures test FEVS at mean PM2.5, not FEVS x PM2.5.
Preserved	Frozen N=97, settled exclusions, all completed imaging results, and original PDF versions.

Conclusion

The current coherent description is provisional financial-vulnerability associations with socially differentiated brain responses. Activation is qualified; the VS-connectivity sensitivity result is more consistent. The environmental-support behavioral link awaits corrected MSPSS scoring.

Native FEAT Design Views: Outcome-Phase Check

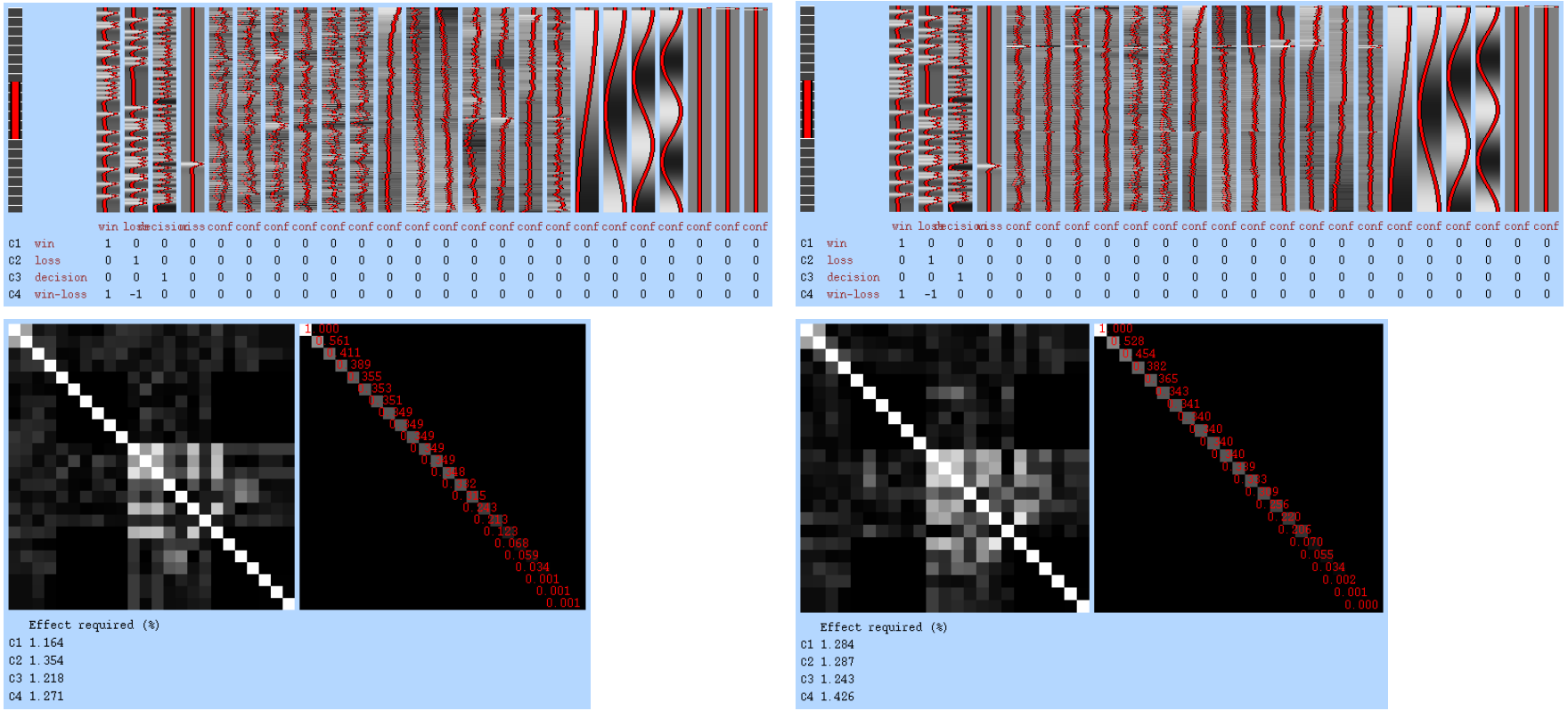
PRELIMINARY | Frozen N=97 | Exploratory, not preregistered

Context

Are the reward estimates tied to the intended task model? These are unmodified native FEAT screenshots from the first frozen participant's two activation L1s, not custom group-design heatmaps.

Content

Social Doors	Nonsocial Doors



Top: native design.png (design and contrast display). Bottom: native design_cov.png. Both tasks' actual design.fsf/.mat/.con were checked for the expected win, loss and win-minus-loss contrast weights before these PNGs were copied unchanged. These are representative design views, not evidence that every participant has identical event timing.

Conclusion

The expected outcome contrasts are present, and all requested evidence artifacts passed the saved-input checks. N=97, previous analyses and exclusions remain unchanged. The remaining measurement issue is scoring verification and the planned comparison of MSPSS administrations, not a new imaging exclusion.